

The Black-Scholes-Merton Model

The Stock Price Assumption

- Consider a stock whose price is S
- In a short period of time of length Δt , the return on the stock is normally distributed:

$$\frac{\Delta S}{S} \approx \phi(\mu \Delta t, \sigma^2 \Delta t)$$

where μ is expected return and σ is volatility

The Lognormal Property

- It follows from this assumption that

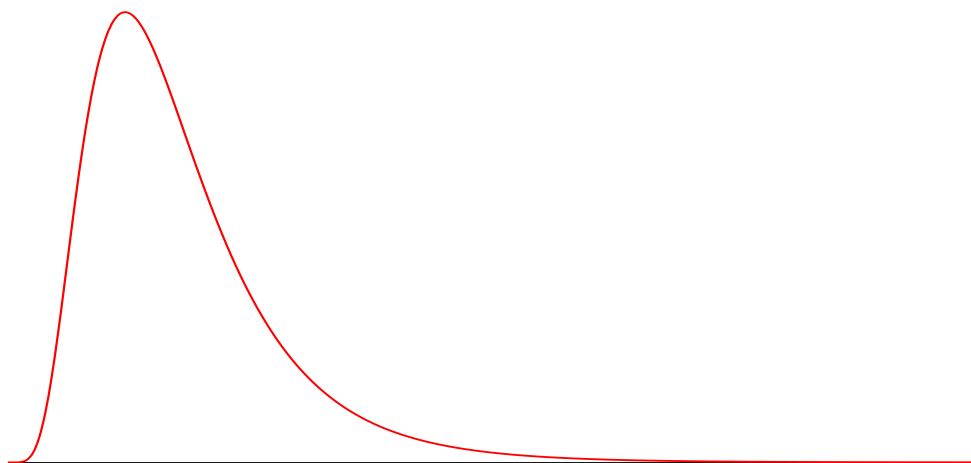
$$\ln S_T - \ln S_0 \approx \phi \left[\left(\mu - \frac{\sigma^2}{2} \right) T, \sigma^2 T \right]$$

or

$$\ln S_T \approx \phi \left[\ln S_0 + \left(\mu - \frac{\sigma^2}{2} \right) T, \sigma^2 T \right]$$

- Since the logarithm of S_T is normal, S_T is lognormally distributed

The Lognormal Distribution



$$E(S_T) = S_0 e^{\mu T}$$

$$\text{var}(S_T) = S_0^2 e^{2\mu T} (e^{\sigma^2 T} - 1)$$

Continuously Compounded Return

If x is the realized continuously compounded return

$$S_T = S_0 e^{xT}$$

$$x = \frac{1}{T} \ln \frac{S_T}{S_0}$$

$$x \approx \phi \left(\mu - \frac{\sigma^2}{2}, \frac{\sigma^2}{T} \right)$$

The Expected Return

- The expected value of the stock price is $S_0 e^{\mu T}$
- The expected return on the stock is

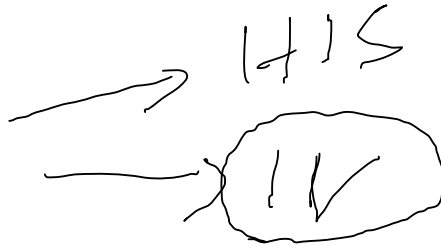
$\mu - \sigma^2/2$ not μ

This is because

$$\ln[E(S_T / S_0)] \quad \text{and} \quad E[\ln(S_T / S_0)]$$

are not the same

The Volatility



- The volatility is the standard deviation of the continuously compounded rate of return in 1 year
- The standard deviation of the return in a short time period time Δt is approximately

$$\underline{\sigma\sqrt{\Delta t}}$$

- If a stock price is \$50 and its volatility is 25% per year what is the standard deviation of the price change in one day?

Estimating Volatility from Historical Data

1. Take observations $S_0, S_1, \dots, S_n$ at intervals of τ years (e.g. for weekly data $\tau = 1/52$)
2. Calculate the continuously compounded return in each interval as:

$$u_i = \ln\left(\frac{S_i}{S_{i-1}}\right)$$

3. Calculate the standard deviation, s , of the u_i 's
4. The historical volatility estimate is: $\hat{\sigma} = \frac{s}{\sqrt{\tau}}$

Nature of Volatility

- Volatility is usually much greater when the market is open (i.e. the asset is trading) than when it is closed
- For this reason time is usually measured in “trading days” not calendar days when options are valued
- It is assumed that there are 252 trading days in one year for most assets

Example

- Suppose it is April 1 and an option lasts to April 30 so that the number of days remaining is 30 calendar days or 22 trading days
- The time to maturity would be assumed to be $22/252 = 0.0873$ years

The Concepts Underlying Black-Scholes-Merton

- The option price and the stock price depend on the same underlying source of uncertainty
- We can form a portfolio consisting of the stock and the option which eliminates this source of uncertainty
- The portfolio is instantaneously riskless and must instantaneously earn the risk-free rate
- This leads to the Black-Scholes-Merton differential equation

The Derivation of the Black-Scholes-Merton Differential Equation

$$\Delta S = \mu S \Delta t + \sigma S \Delta z$$

$$\Delta f = \left(\frac{\partial f}{\partial S} \mu S + \frac{\partial f}{\partial t} + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 S^2 \right) \Delta t + \frac{\partial f}{\partial S} \sigma S \Delta z$$

We set up a portfolio consisting of

– 1: derivative

+ $\frac{\partial f}{\partial S}$: shares

This gets rid of the dependence on Δz .

The Derivation of the Black-Scholes-Merton Differential Equation

The value of the portfolio, Π , is given by

$$\Pi = -f + \frac{\partial f}{\partial S} S$$

The change in its value in time Δt is given by

$$\Delta \Pi = -\Delta f + \frac{\partial f}{\partial S} \Delta S$$

The Derivation of the Black-Scholes-Merton Differential Equation continued

The return on the portfolio must be the risk - free rate. Hence

$$\Delta \Pi = r \Pi \Delta t$$

$$-\Delta f + \frac{\partial f}{\partial S} \Delta S = r \left(-f + \frac{\partial f}{\partial S} S \right) \Delta t$$

We substitute for Δf and ΔS in this equation to get the Black - Scholes differential equation :

$$\frac{\partial f}{\partial t} + rS \frac{\partial f}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 f}{\partial S^2} = rf$$

The Differential Equation

- Any security whose price is dependent on the stock price satisfies the differential equation
- The particular security being valued is determined by the boundary conditions of the differential equation.
- The boundary condition for European call option is

$$f = \max(S - K, 0) \quad \text{when } t = T$$

The boundary condition for European put option is

$$f = \max(K - S, 0) \quad \text{when } t = T$$

The Black-Scholes-Merton Formulas for Options

$$c = S_0 N(d_1) - K e^{-rT} N(d_2)$$

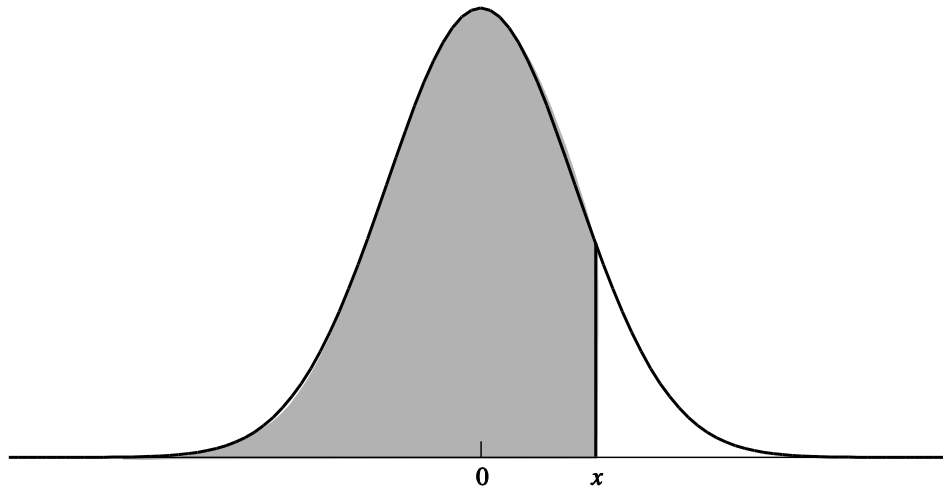
$$p = K e^{-rT} N(-d_2) - S_0 N(-d_1)$$

$$\text{where } d_1 = \frac{\ln(S_0 / K) + (r + \sigma^2 / 2)T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{\ln(S_0 / K) + (r - \sigma^2 / 2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

The $N(x)$ Function

- $N(x)$ is the probability that a normally distributed variable with a mean of zero and a standard deviation of 1 is less than x



Properties of Black-Scholes Formula

- As S_0 becomes very large c tends to $S_0 - Ke^{-rT}$ and p tends to zero
- As S_0 becomes very small c tends to zero and p tends to $Ke^{-rT} - S_0$
- What happens as σ becomes very large?
- What happens as T becomes very large?

Understanding Black-Scholes

$$c = e^{-rT} N(d_2) \left(S_0 e^{rT} N(d_1) / N(d_2) - K \right)$$

e^{-rT} : Present value factor

$N(d_2)$: Probability of exercise

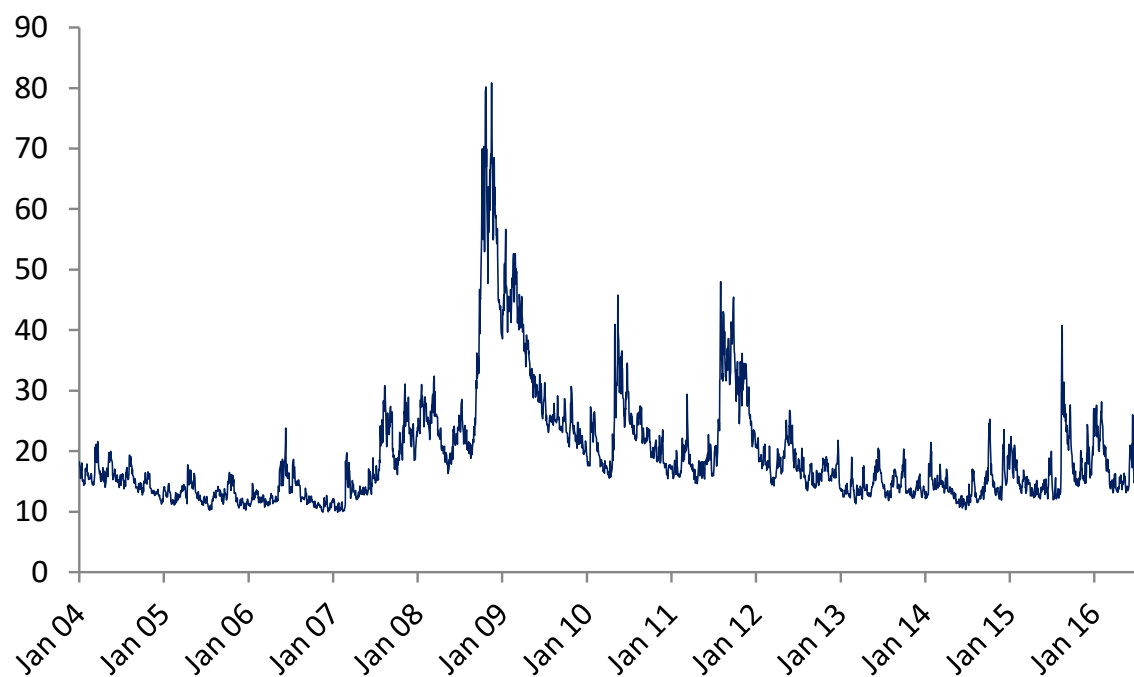
$S_0 e^{rT} N(d_1) / N(d_2)$: Expected stock price in a risk - neutral world
if option is exercised

K : Strike price paid if option is exercised

Implied Volatility

- The implied volatility of an option is the volatility for which the Black-Scholes-Merton price equals the market price
- There is a one-to-one correspondence between prices and implied volatilities
- Traders and brokers often quote implied volatilities rather than dollar prices

The VIX S&P500 Volatility Index



An Issue of Warrants & Executive Stock Options

- When a regular call option is exercised the stock that is delivered must be purchased in the open market
- When a warrant or executive stock option is exercised new Treasury stock is issued by the company
- If little or no benefits are foreseen by the market, the stock price will reduce at the time the issue is announced.

The Impact of Dilution

- After the options have been issued it is not necessary to take account of dilution when they are valued
- Before they are issued we can calculate the cost of each option as $N/(N+M)$ times the price of a regular option with the same terms where N is the number of existing shares and M is the number of new shares that will be created if exercise takes place

Dividends

- European options on dividend-paying stocks are valued by substituting the stock price less the present value of dividends into Black-Scholes
- Only dividends with ex-dividend dates during life of option should be included
- The “dividend” should be the expected reduction in the stock price expected